

Project Report: Multi-User Distributed OS-Shell

Course: EGC 301P – Operating Systems Lab

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1. Problem Statement

The goal of this project is to implement a command-line interpreter (shell) in C that demonstrates core Operating Systems principles. The system provides a secure environment where multiple users can execute local commands, manage background jobs, and interact with a remote shell server through a custom TCP-based protocol.

2. Implementation of Required Concepts

2.1 Role-Based Authorization (Section 4.1)

The shell implements a robust security layer with three roles: **Admin**, **User**, and **Guest**.

- **Implementation:** As seen in your third screenshot, when logged in as a **Guest**, attempting to run restricted commands (e.g., `cat README.md` via a typo or unauthorized access) results in a "**Permission denied**" message.
- **Access Control:** The `is_allowed()` function checks the `current_role` against a command whitelist/blacklist before execution.

2.2 Concurrency Control & Data Consistency (Sections 4.3 & 4.4)

The system handles multiple simultaneous processes and protects shared data.

- **Mutex Synchronization:** A `pthread_mutex_t jobs_lock` is used to prevent race conditions when the shell's main thread and the `SIGCHLD` signal handler attempt to modify the background jobs array at the same time.
- **Demonstration:** Your first two screenshots show the shell successfully managing multiple concurrent background tasks (`sleep 60 &`) without data corruption.

2.3 Socket Programming (Section 4.5)

The project utilizes a **Client-Server model** for remote interaction.

- **Implementation:** Launching the shell with the `--server` flag starts a TCP listener on port 9090.
- **Remote Execution:** As shown in the side-by-side terminal screenshot, a client uses the `rsh` command to send an `ls` request to the server, which then executes the command and returns the output over the network.

2.4 Inter-Process Communication (IPC) (Section 4.6)

- **Pipes:** The system supports multi-stage command chaining. Your fourth screenshot demonstrates a three-stage pipeline: `ls | grep ".c" | wc -l`.
- **Signals:** The fifth screenshot shows the implementation of SIGTSTP (Ctrl+Z). When the user sends a stop signal to a foreground process (`sleep 100`), the shell captures it, moves the process to a **"Stopped"** state, and adds it to the job list.

2.5 File Locking (Section 4.2)

- **Implementation:** The shell uses `fcntl` advisory locking to manage the `history.log` file. This prevents interleaving or corruption of log entries when multiple shell instances write to the history simultaneously.

3. Screenshots and Output Analysis

Feature	Screenshot Evidence
Compilation	Using <code>make</code> to compile with <code>-lpthread</code> for concurrency support.
Job Management	Successful execution of five background <code>sleep</code> processes.
Access Control	Guest login failing to access files with a "Permission denied" error.
IPC (Pipes)	Successful execution of a complex pipeline for file counting.
Signal Handling	Moving a process to a Stopped state via Ctrl+Z.
Remote Shell	Side-by-side view of a server listening and a client executing <code>rsh</code> .

4. Challenges Faced and Solutions

Challenge 1: Path Resolution during Directory Traversal

- **Issue:** Using `cd` caused the shell to lose access to `data/users.txt` because relative paths were no longer valid.
- **Solution:** Implemented a global `BASE_DIR` anchor using `getcwd()` at startup, ensuring all file operations are relative to the project root regardless of the current working directory.

Challenge 2: Interrupted System Calls

- **Issue:** Signal handlers for `SIGCHLD` and `SIGTSTP` were causing `fgets()` to return early with an error, freezing the shell.
- **Solution:** Implemented `EINTR` error checking in the main REPL loop to automatically resume the prompt after a signal interruption, maintaining system responsiveness.

```
jaswanth@jaswanths-MacBook-Air OS-SHELL % gcc -Iinclude src/main.c src/shell.c src/server.c src/client.c -o oshell
jaswanth@jaswanths-MacBook-Air OS-SHELL % ./oshell
Username: admin
Password: admin
Login successful as admin
admin@oshell> sleep 60 &
[1] 24376
admin@oshell> sleep 60 &
[2] 24378
admin@oshell> sleep 60 &
[3] 24379
admin@oshell> sleep 60 &
[4] 24381
admin@oshell> sleep 60 &
[5] 24382
admin@oshell> sleep 60 &
█

jaswanth@jaswanths-MacBook-Air OS-SHELL % gcc -Iinclude src/main.c src/shell.c src/server.c src/client.c -o oshell
jaswanth@jaswanths-MacBook-Air OS-SHELL % ./oshell
Username: admin
Password: admin
Login successful as admin
admin@oshell> sleep 60 &
[1] 24376
admin@oshell> sleep 60 &
[2] 24378
admin@oshell> sleep 60 &
[3] 24379
admin@oshell> sleep 60 &
[4] 24381
admin@oshell> sleep 60 &
[5] 24382
admin@oshell> sleep 60 &
[5] 24383
admin@oshell> █
```

```

Problems 20 Output Debug Console Terminal Ports
● jaswanth@jaswanths-MacBook-Air OS-SHELL % make
gcc -Wall -Wextra -I include -o oshell src/main.c src/shell.c
src/server.c src/client.c -lpthread
● jaswanth@jaswanths-MacBook-Air OS-SHELL % ./oshell
Username: admin
Password: admin
Login successful as admin
admin@oshell> exit
● jaswanth@jaswanths-MacBook-Air OS-SHELL % clear
○ jaswanth@jaswanths-MacBook-Air OS-SHELL % ./oshell --server
Server listening on port 9090...
[ ]
○ jaswanth@jaswanths-MacBook-Air OS-SHELL % ./oshell
Username: admin
Password: admin
Login successful as admin
admin@oshell> rsh 127.0.0.1 9090 ls
data
include
Makefile
oshell
PROJECT_REPORT.md
PROJECT_REPORT.txt
Project-Report.pdf
README.md
src
tests
admin@oshell> history
sleep 60 &
sleep 60 &
sleep 60 &
sleep 60 &
sleep 60 &
sleep 60 &
exit
ls
car README.md
ls | grep ".c" | wc -l
exit
ls | grep ".c" | wc -l
exit
sleep 100
exit
sleep 100
jobs
exit

```

```

● jaswanth@jaswanths-MacBook-Air OS-SHELL % make
gcc -Wall -Wextra -I include -o oshell src/main.c src/shell.c src/server.c src/client.c -lpthread
○ jaswanth@jaswanths-MacBook-Air OS-SHELL % ./oshell
Username: admin
Password: admin
Login successful as admin
admin@oshell> sleep 100
^Z
[1]+ Stopped sleep
admin@oshell> jobs
[1] Running sleep [25726]
admin@oshell>

```

```

● jaswanth@jaswanths-MacBook-Air OS-SHELL % make
gcc -Wall -Wextra -I include -o oshell src/main.c src/shell.c src/server.c src/client.c -lpthread
○ jaswanth@jaswanths-MacBook-Air OS-SHELL % ./oshell
Username: admin
Password: admin
Login successful as admin
admin@oshell> ls | grep ".c" | wc -l
0
admin@oshell>

```

```

● jaswanth@jaswanths-MacBook-Air OS-SHELL % make
gcc -Wall -Wextra -I include -o oshell src/main.c src/shell.c src/server.c src/client.c -lpthread
○ jaswanth@jaswanths-MacBook-Air OS-SHELL % ./oshell
Username: guest
Password: guest
Login successful as guest
guest@oshell> ls
data                Makefile            PROJECT_REPORT.md   Project-Report.pdf  src
include             oshell             PROJECT_REPORT.txt  README.md            tests
guest@oshell> car README.md
Permission denied
guest@oshell>

```